

# **UCS615**

## **MICROPROCESSOR BASED SYSTEM DESIGN**



**THAPAR INSTITUTE**  
OF ENGINEERING & TECHNOLOGY  
(Deemed to be University)

### **LAB MANUAL**

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# Experiment 1

## Objective

To write and execute an assembly language program using an 8086 microprocessor that adds two 16-bit numbers.

## Theory

The Intel 8086 microprocessor is a 16-bit processor which allows arithmetic operations like addition, subtraction, multiplication, and division. The `ADD` instruction is used to perform addition of two registers or a register and memory. When two 16-bit numbers are added, the result may or may not generate a carry which is stored in the carry flag (CF).

The 8086 architecture uses registers like AX, BX, CX, DX (general purpose registers) to perform such operations.

## Algorithm

The following is the step-by-step procedure followed to develop the assembly language program and achieve the experiment's objective.

1. Start the program.
2. Load the first 16-bit number into register AX.
3. Load the second 16-bit number into register BX.
4. Add the contents of BX to AX using the `ADD` instruction.
5. Store or observe the result in AX.
6. Terminate the program using `HLT`.

## Program

The following is the assembly language program written to add two 16-bit numbers using the 8086 microprocessor.

```
MOV AX, 1234H ; Load first number into AX
MOV BX, 1236H ; Load second number into BX
ADD AX, BX    ; Add contents of BX to AX
HLT           ; Halt the program
```

## Output

The output of the program will be the sum of the two 16-bit numbers stored in register AX. For the given example, the result will be: `AX = 246AH`

## Precautions

1. Ensure that the numbers being added do not exceed the 16-bit limit (0 to 65535).
2. Always check the carry flag (CF) after addition to handle overflow situations.
3. Use appropriate registers for storing intermediate results to avoid data loss.
4. Test the program with different sets of numbers to ensure accuracy.